EXPERIMENT – 4

AIM: To perform and analyze SQL queries using INNER JOIN, LEFT JOIN, SELF JOIN, 3way JOIN, correlated subqueries, EXISTS, and simulated INTERSECT and EXCEPT operations on the Employee–Department–Project database, and to compare their execution plans using EXPLAIN.

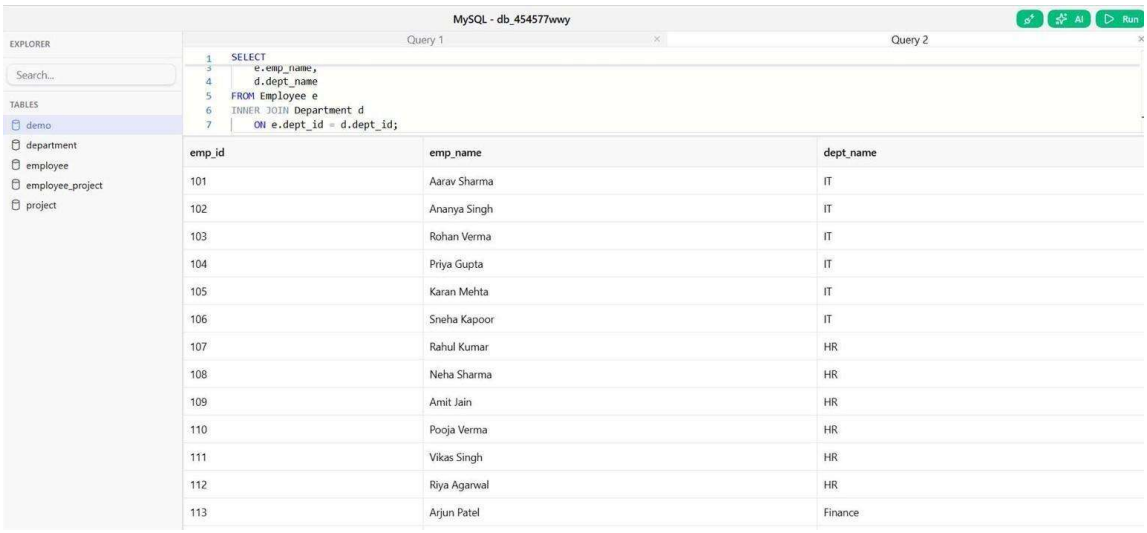
SOURCE CODE: USING EXPERIMENT – 3 DATABASE

SQL QUERIES:

INNER JOIN

```
SELECT
        e.emp_id,
        e.emp_name,
        d.dept_name
FROM Employee e
INNER JOIN Department d
        ON e.dept_id = d.dept_id;
```

OUTPUT:

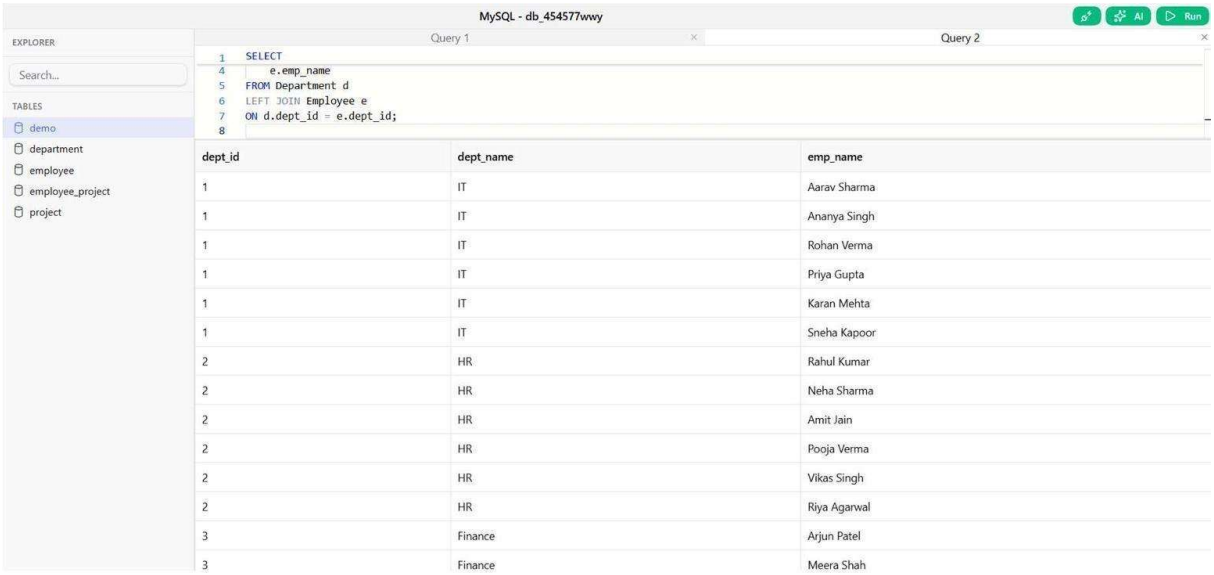


emp_id	emp_name	dept_name
101	Aarav Sharma	IT
102	Ananya Singh	IT
103	Rohan Verma	IT
104	Priya Gupta	IT
105	Karan Mehta	IT
106	Sneha Kapoor	IT
107	Rahul Kumar	HR
108	Neha Sharma	HR
109	Amit Jain	HR
110	Pooja Verma	HR
111	Vikas Singh	HR
112	Riya Agarwal	HR
113	Arjun Patel	Finance

LEFT JOIN

```
SELECT
        d.dept_id,
        d.dept_name,
        e.emp_name
FROM Department d
LEFT JOIN Employee e
        ON d.dept_id = e.dept_id;
```

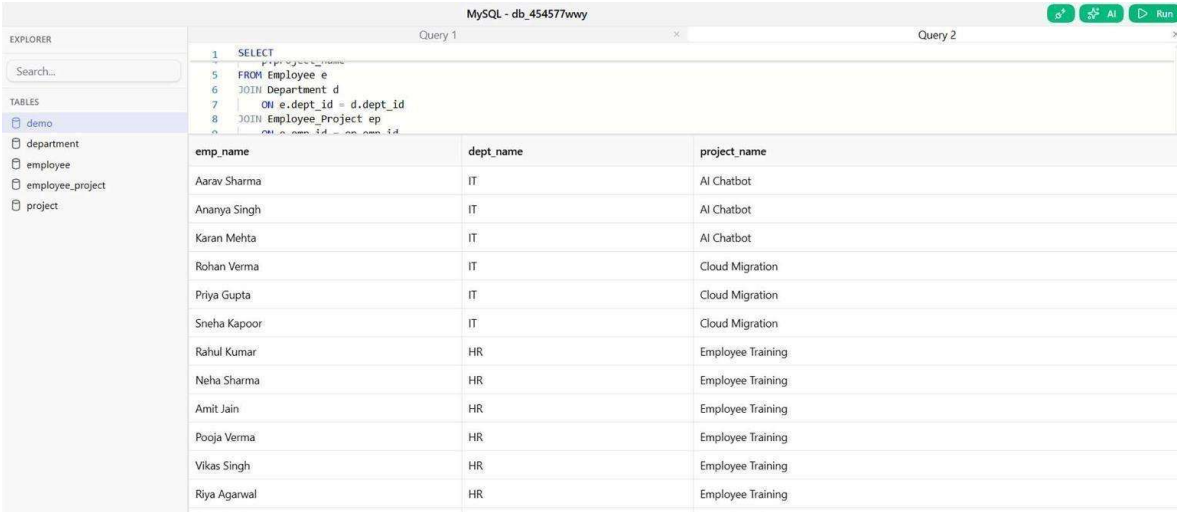
OUTPUT:



3-WAY JOIN

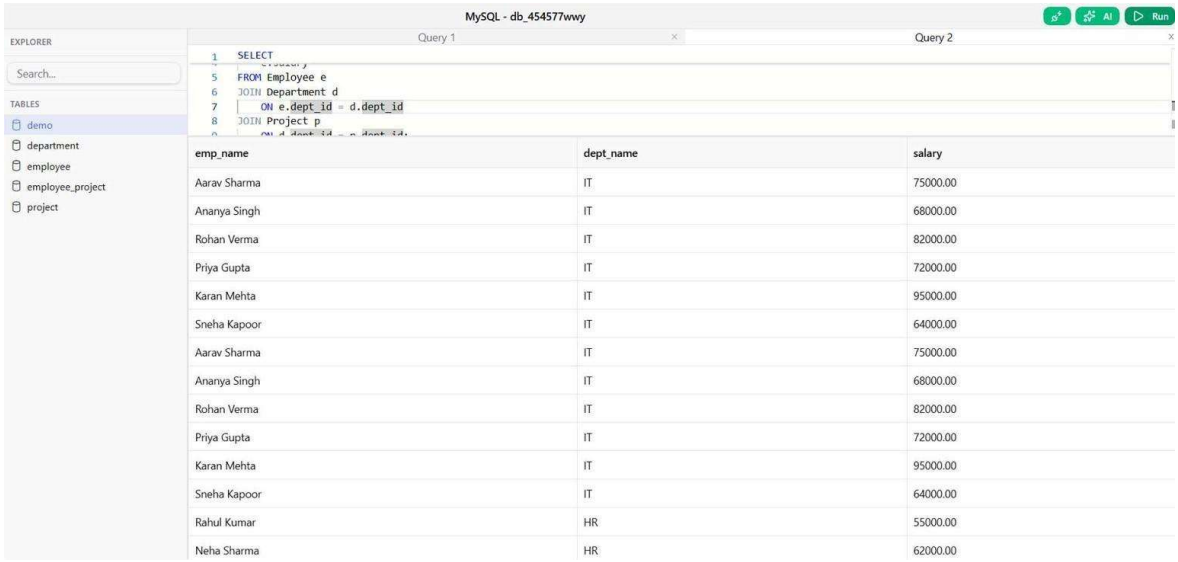
```
SELECT
        e.emp_name,
        d.dept_name,
        p.project_name
FROM Employee e
JOIN Department d
        ON e.dept_id = d.dept_id
JOIN Employee_Project ep
        ON e.emp_id = ep.emp_id
JOIN Project p
        ON ep.project_id = p.project_id;
```

OUTPUT:



```
SELECT
        e.emp_name,
        d.dept_name,
        e.salary
FROM Employee e
JOIN Department d
        ON e.dept_id = d.dept_id
JOIN Project p
        ON d.dept_id = p.dept_id;
```

OUTPUT:

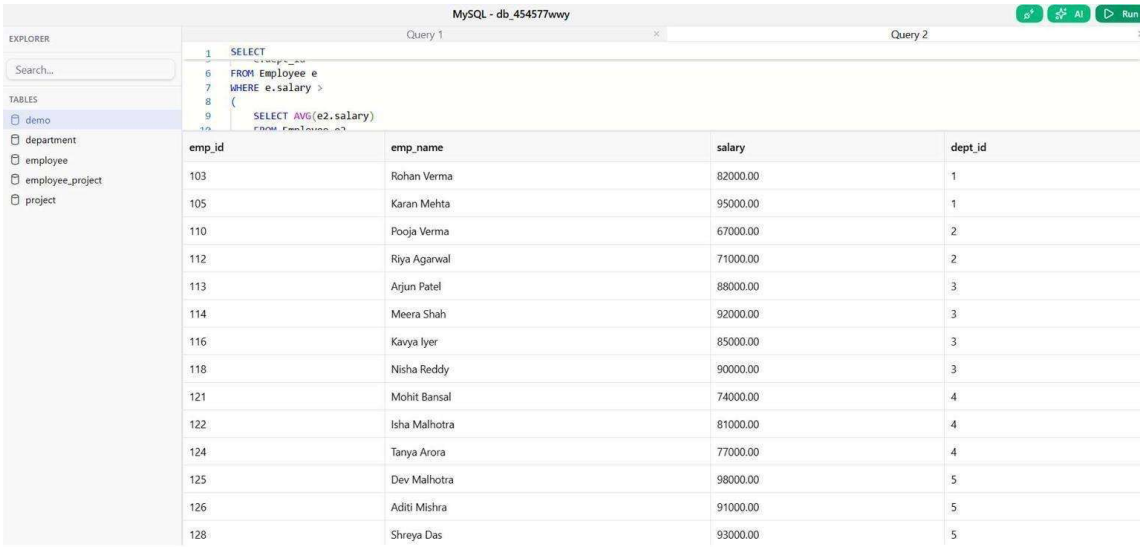


emp_name	dept_name	salary
Aarav Sharma	IT	75000.00
Ananya Singh	IT	68000.00
Rohan Verma	IT	82000.00
Priya Gupta	IT	72000.00
Karan Mehta	IT	95000.00
Sneha Kapoor	IT	64000.00
Aarav Sharma	IT	75000.00
Ananya Singh	IT	68000.00
Rohan Verma	IT	82000.00
Priya Gupta	IT	72000.00
Karan Mehta	IT	95000.00
Sneha Kapoor	IT	64000.00
Rahul Kumar	HR	55000.00
Neha Sharma	HR	62000.00

CORRELATED SUBQUERY

```
SELECT
        e.emp_id,  e.emp_name,  e.salary,  e.dept_id
FROM Employee e
WHERE e.salary >
(
        SELECT AVG(e2.salary)
        FROM Employee e2
        WHERE e2.dept_id = e.dept_id
);
```

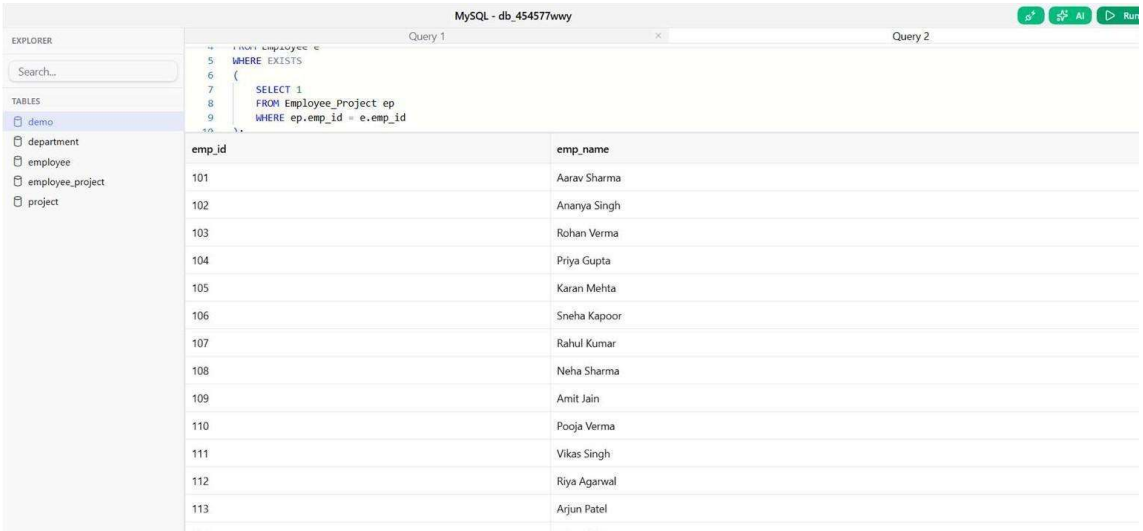
OUTPUT:



EXISTS

```
SELECT
        e.emp_id,
        e.emp_name
FROM Employee e
WHERE EXISTS
(
        SELECT 1
        FROM Employee_Project ep
        WHERE ep.emp_id = e.emp_id
);
```

OUTPUT:



SIMULATED INTERSECT

```
SELECT
        e.emp_id,
        e.emp_name
FROM Employee e
```

```
WHERE e.dept_id = 1
AND EXISTS
(
        SELECT 1
        FROM Employee_Project ep
        WHERE ep.emp_id = e.emp_id
                AND ep.project_id = 201
);
```

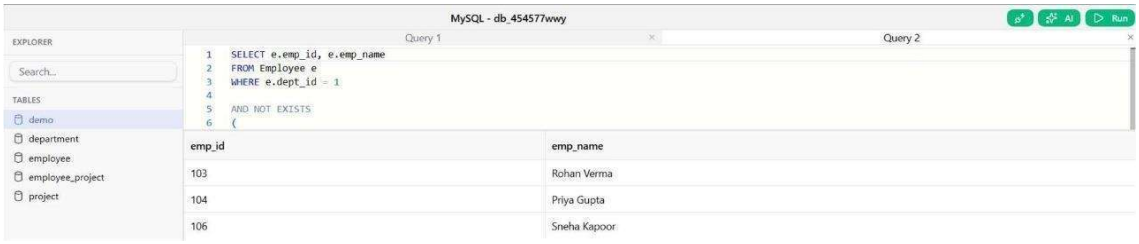
OUTPUT:



SIMULATED EXCEPT

```
SELECT e.emp_id, e.emp_name FROM Employee e
WHERE e.dept_id = 1
AND NOT EXISTS (
SELECT 1
FROM Employee_Project ep WHERE ep.emp_id = e.emp_id
AND ep.project_id = 201
);
```

OUTPUT:

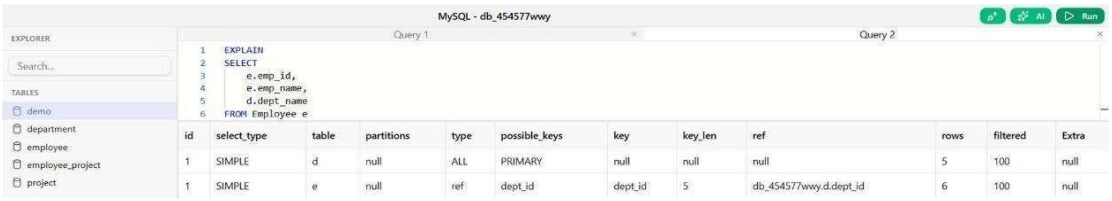


COMPARING EXECUTION PLANS USING EXPLAIN EXPLAIN – INNER JOIN

EXPLAIN

```
SELECT
e.emp_id,
e.emp_name,
d.dept_name
FROM Employee e
INNER JOIN Department d
ON e.dept_id = d.dept_id;
```

OUTPUT:

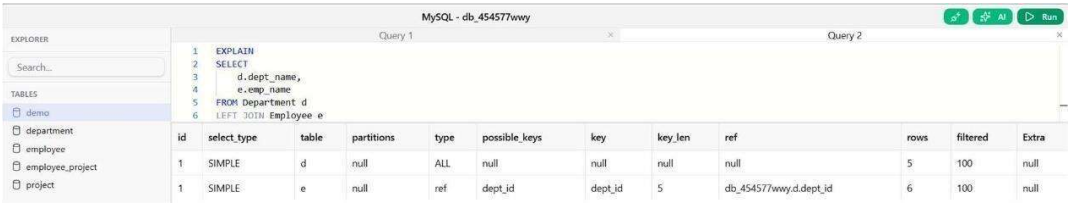


EXPLAIN – LEFT JOIN

EXPLAIN

SELECT
d.dept_name, e.emp_name
FROM Department d LEFT JOIN Employee e
ON d.dept_id = e.dept_id;

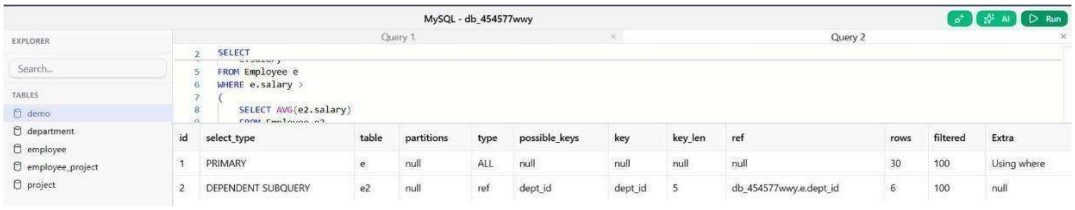
OUTPUT:



EXPLAIN – CORRELATED SUBQUERY

```
EXPLAIN SELECT
        e.emp_name, e.salary
FROM Employee e
WHERE e.salary >
(
        SELECT AVG(e2.salary)
        FROM Employee e2
        WHERE e2.dept_id = e.dept_id
)
```

OUTPUT:



EXPLAIN – EXISTS

```
EXPLAIN SELECT
e.emp_id, e.emp_name
FROM Employee e
WHERE EXISTS
(
SELECT 1
FROM Employee_Project ep
WHERE ep.emp_id = e.emp_id
);
```

OUTPUT:

